Transformer and CNN Hybrid Deep Neural Network for Semantic Segmentation of Very-High-Resolution Remote Sensing Imagery

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***Abstract*— This article presents a transformer and convolutional neural network (CNN) hybrid deep neural network for semantic segmentation of very high resolution (VHR) remote sensing imagery. The model follows an encoder–decoder structure. The encoder module uses a new universal backbone Swin transformer to extract features to achieve better long-range spatial dependencies modeling. The decoder module draws on some effective blocks and successful strategies of CNN-based models in remote sensing image segmentation. In the middle of the framework, an atrous spatial pyramid pooling block based on depthwise separable convolution (SASPP) is applied to obtain a multiscale context. A U-shaped decoder is used to gradually restore the size of the feature maps. Three skip connections are built between the encoder and decoder feature maps of the same size to maintain the transmission of local details and enhance the communication of multiscale features. A squeeze-and-excitation (SE) channel attention block is added before segmentation for feature augmentation. An auxiliary boundary detection branch is combined to provide edge constraints for semantic segmentation. Extensive ablation experiments were conducted on the International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen and Potsdam benchmarks to test the effectiveness of multiple components of the network. At the same time, the proposed method is compared with the current state-of-the-art methods on the two benchmarks. The proposed hybrid network achieved the second highest overall accuracy (OA) on both the Potsdam and Vaihingen benchmarks (code and models are available at https://github.com/zq7734509/mmsegmentation-multilayer).**

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***Index Terms*— Boundary detection, semantic segmentation, squeeze-and-excitation (SE) block, Swin transformer, very high resolution (VHR) remote sensing imagery.**

I. Introduction

SEMANTIC segmentation is to assign semantic labels to each pixel of an image. In the field of remote sensing, semantic segmentation is also called the classification of land use and land cover (LULC) types. As basic research of urban remote sensing image analysis, it provides support for many important remote sensing applications, such as geographic information database update, urban planning, and urban LULC change detection.

Remote sensing technology can provide a rich source of data for Earth observation. With the development of sensor technology, a large number of very high resolution (VHR) remote sensing images can be obtained [1], [2]. While meeting the needs of more basic surveying and mapping applications, it also challenges the capabilities of semantic segmentation algorithms. On the one hand, a VHR image can capture detailed features, such as the internal structure, texture, and color characteristics of targets, so it shows the characteristics of large intraclass variance and small interclass variance, especially in urban scenes. On the other hand, many artificial targets in urban scenes may produce umbra and falling shadows, which may affect the spectral values of pixels in the image. The above two reasons will greatly interfere with the correct interpretation of pixels in the VHR image.

In recent years, deep convolutional neural networks (DCNN) have become the state-of-the-art image analysis technology. DCNN can provide end-to-end solutions for multiple visual tasks, such as instance segmentation [3]–[5], semantic segmentation [6]–[9], and object detection [10], [11]. They take raw images as input and outputs the prediction results directly, without the need for feature engineering. Multiple sets of stacked convolution and pooling operations give DCNN powerful multiscale context extraction capabilities. In the field of remote sensing, DCNN breaks the accuracy bottleneck of traditional object- or pixel-oriented segmentation methods and also improves the automation level of VHR remote sensing image interpretation.

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In the last few years, many works have tried various convolutional neural network (CNN)-based backbones in the application of remote sensing image semantic segmentation, such as VGGNet [12], ResNet [13], and HRNet [14]. Related research includes increasing the width and depth of the network, the interaction of local and global information, attention mechanism, multiscale feature expression, and so on. The DCNN architectures have specific inductive deviations, including locality and translation invariance. This is determined by the local connection (kernel) and weight sharing of the convolution operation. The two strategies effectively reduce the number of parameters and improve the learning efficiency of the CNN-based encoders. However, the inductive deviations may also limit the model's extraction ability of long-range spatial dependence and lead to the reduction of the model's performance.

Transformer is a concept proposed in 2017 [15]. It is a structure that learns features through a self-attention mechanism. The transformer was first applied to the field of natural language and achieved outstanding performance. At present, the transformer has been considered as the preferred model for natural language processing tasks. Inspired by this, researchers recently applied transformers to computer vision tasks. For the first time, the vision transformer (ViT) uses a pure transformer structure as a feature extractor to process image recognition task [16]. Under the condition of being pretrained on a large-scale dataset, ViT achieved slightly better accuracy than the CNN-based structure. This proves the powerful feature extraction capabilities of the transformer model and its potential in the field of image processing. Subsequently, more researchers proposed transformer-based models to deal with various visual tasks, such as object detection [17], semantic segmentation [18], and video understanding [19]. The segmentation transformer (SETR) [18] is the first semantic segmentation network that uses a pure transformer structure as backbone. It designed three CNN-based decoders for the transformer-based encoder and achieves the best accuracy on the ADE20K (50.28% mIoU) and Pascal Context datasets.

Transformer structures, such as ViT and SETR, convert the original image into sequence-to-sequence image blocks as inputs. Despite their powerful feature extraction capabilities, they also require a lot of memory and computing resources. The computational complexity of self-attention increases with the size of the image as a square multiple, which makes it difficult for the transformer structure based on sequence-to-sequence image blocks to process high-resolution images. Moreover, the size of the image block is fixed, and the backbone can only output a single-scale low-resolution feature map [the original setting of SETR is $(1/16)$], which is not suitable for processing densely labeled semantic segmentation task.

In order to reduce the computational complexity of the transformer-based backbone and generate multiscale feature maps for better versatility, the Swin transformer proposes a hierarchical transformer structure [20]. The backbone limits the self-attention calculation to nonoverlapping partial windows and establishes cross-window connections through window shift, which greatly improves the efficiency of the self-attention calculation. In addition, the Swin transformer structure can output hierarchical feature maps, which can be perfectly integrated into the existing encoder–decoder structure.

In this article, we apply the Swin transformer structure to the task of semantic segmentation of VHR remote sensing images, design a CNN-based decoder to recover the size of the feature maps, and obtain the semantic segmentation results. In the decoder module, multiple effective blocks and successful strategies in the research of CNN-based VHR remote sensing images are applied. The main contributions of this article are given as follows.

1) We combined the Swin transformer backbone with the CNN-based decoder to build a new deep learning framework for semantic segmentation of VHR remote sensing imagery. The structure of the Swin transformer enables it to model the long-range spatial dependence in the image and obtain hierarchical features simultaneously.
2) A U-shaped decoder that integrates skip connections, the atrous spatial pyramid pooling block based on the depthwise separable convolution (SASPP) block, and SE block is designed to preserve local details, restore the resolution of feature maps, and obtain competitive segmentation results. Through ablation studies, we tested the effect of each component in the decoder module.
3) We comprehensively compared the effects of several most popular CNN- and transformer-based backbones for semantic segmentation of VHR remote sensing imagery, and further visualized and analyzed the characteristics of their output feature maps.
4) Inspired by the CNN-based remote sensing image analysis, we adopt a dual-task network design and added an auxiliary boundary detection task to improve the accuracy of boundary pixel positioning and reduce the salt and pepper noise for semantic segmentation. Considering the difference between the feature maps extracted by the CNN-based backbone and the transformer-based backbone, we designed two boundary detection strategies. Through experimental comparisons, we chose a more effective boundary detection branch in the proposed network.

II. Related Works

A. Review of FCN-Based Semantic Segmentation Methods

The fully convolutional network (FCN) is a milestone in CNN-based semantic segmentation methods [21]. It replaces the fully connected layers in the Visual Geometry Group (VGG) structure with convolutional layers to establish an end-to-end semantic segmentation network. The FCN architecture also proposes other two innovations. First, an encoder–decoder framework is proposed to restore the resolution of feature maps. Second, the work proposes to use skip connections to establish an information flow between deep and shallow features. Therefore, the detailed spatial information will not be lost after the pooling operation.

In the standard encoder–decoder structure, the encoder module is used for feature extraction and has received more attention. Hierarchical feature maps are obtained through stacked convolution, activation, and pooling layers. The most popular

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encoders include VGGNet [12], ResNet [13], HRNet [14], ResNeSt [22], and so on. ResNet [13] proposes the residual structure by introducing identity shortcut connection to deal with the problem of vanishing gradients, which provides the condition for training deeper CNN models. ResNeXt [23] proposes to replace the residual blocks of the ResNet with grouped convolutions, thereby increasing the width of the network and reducing the number of model parameters. ResNeSt [22] combines the squeeze-and-excitation (SE) attention block on the basis of the ResNeXt's cardinality to enhance the interaction of channel information. The abovementioned networks all extract high-level semantic information from low-resolution feature maps. In order to maintain high-resolution expression during the feature extraction process, Sun *et al.* [14] proposed the HRNet. Starting from high-resolution features, HRNet gradually generates parallel feature maps from high-to-low resolution at each stage. Through multiple exchanges of information on parallel multiresolution feature maps, the HRNet can obtain repeated multiresolution fusions.

The function of the decoder is to restore the resolution and make full use of the multiscale feature maps to predict semantic segmentation results. At present, the most commonly used decoders are Deeplabv3+ [6] and symmetrical autodecode structures. Deeplabv3+ is the latest version of the Deeplab series. Its encoder and decoder modules both contain novel designs. In the encoder module, atrous convolution is applied to increase the receptive field without reducing the resolution of the feature maps. Multiscale features are extracted using the proposed SASPP block. In the decoder module, Deeplabv3+ builds a skip connection between the low- and high-resolution feature maps to restore detailed information. Meanwhile, depthwise separable convolution is applied to reduce model parameters and computational consumption.

Early influential symmetrical encoder–decoder structures include SegNet [24], DeconvNet [25], U-Net [26], and so on. Variants of U-Net have become the most concerned symmetric encoder–decoder structure due to their outstanding performance in many fields, such as medical and VHR remote sensing image segmentation. The U-shaped decoder gradually restores the resolution of the feature maps through upsampling operations; meanwhile, several skip connections are built between the downsampling and upsampling paths. Therefore, multiscale feature maps can be better fused in the U-shaped decoder, and the output feature maps can retain low-level features to obtain detailed local information. The novel U-shaped decoder usually follows the structure of U-Net and incorporates the most advanced research at the time. For example, inspired by ResNet and DenseNet [27], Res-Unet [28] and Dense-Unet [29] replace each submodule of UNet with the form of residual connection and dense connection, respectively.

The advancement of deep learning methods has promoted their application in the field of remote sensing. Many deep learning solutions for semantic segmentation of remote sensing image have been proposed recently [7]–[9], [30]–[37], and we select some research most related to the work of this article to review. ResUNet [32] proposes an encoder–decoder framework for semantic segmentation of VHR imagery based on the U-Net structure. The encoder combines multiple parallel branches in ResNet to increase the width of the network; a pyramid scene parsing pooling block is applied to obtain multiscale contexts. A variant of the dice loss function is designed to speed up the convergence of the model and improve its performance. The network learns four complementary tasks at the same time, namely, the boundary of the objects, the distance transform of the segmentation mask, the segmentation mask, and a colored reconstruction of the input. Their results are obtained by reasoning with each other. The model that we proposed uses a similar dual-task design, but our two tasks are in parallel rather than inferred by each other. CASIA2 proposes a self-cascaded CNN for segmentation [36]. The backbone is departed from the VGGNet, and multiscale contexts are obtained through a novel self-cascaded manner from coarse to fine. A U-shaped decoder is applied to gradually restore the resolution of the feature map; meanwhile, skip connections are built to maintain low-level features. Attention fused network and test time augmentation (AFNet + TTA) [37] uses a dual-branch residual network structure to process the red green & blue (RGB) images and the normalized digital surface models (nDSM) data separately. In the process of feature map fusion, both spatial attention and channel attention are combined. The decoder module uses a U-shaped structure and adds three skip connections. In addition, spatial and channel attention blocks are also combined before the final prediction of semantic segmentation.

B. Review of Transformer-Based Semantic Segmentation Methods

The CNN model is based on the inductive bias of locality and translation invariance, which makes the model more efficient and easier to train. However, at the same time, the inductive bias may also limit the model's extraction ability of long-range spatial dependence and lead to the reduction of the model's performance. The transformer is a new type of neural network structure that uses self-attention to study the spatial correlation between pixels in the entire image. Before using a pure transformer as the main encoder to extract features, efforts have been done to use self-attention to assist CNN for feature extraction. Most of these models apply self-attention to feature maps output by CNN backbone to enhance the encoder's expression of long-distance spatial dependence. Representative research includes DNLNet [38], Nonlocal [39], DANet [40], CCNet [41], and so on.

Inspired by the great success of transformers in the field of natural language processing, some studies have tried to use a transformer as the backbone instead of CNN to process vision tasks. ViT [16] is groundbreaking research that applies a pure transformer structure to process image recognition tasks. It divides the raw image into a sequence of nonoverlap blocks; the image blocks are then flattened and mapped to D-dimensions through patch embedding. A classification head and a position embedding are attached, and finally, the output is fed into the transformer structure. The performance of this model on medium-scale datasets is slightly lower than that of the ResNet-style CNN structure, but, after being pretrained

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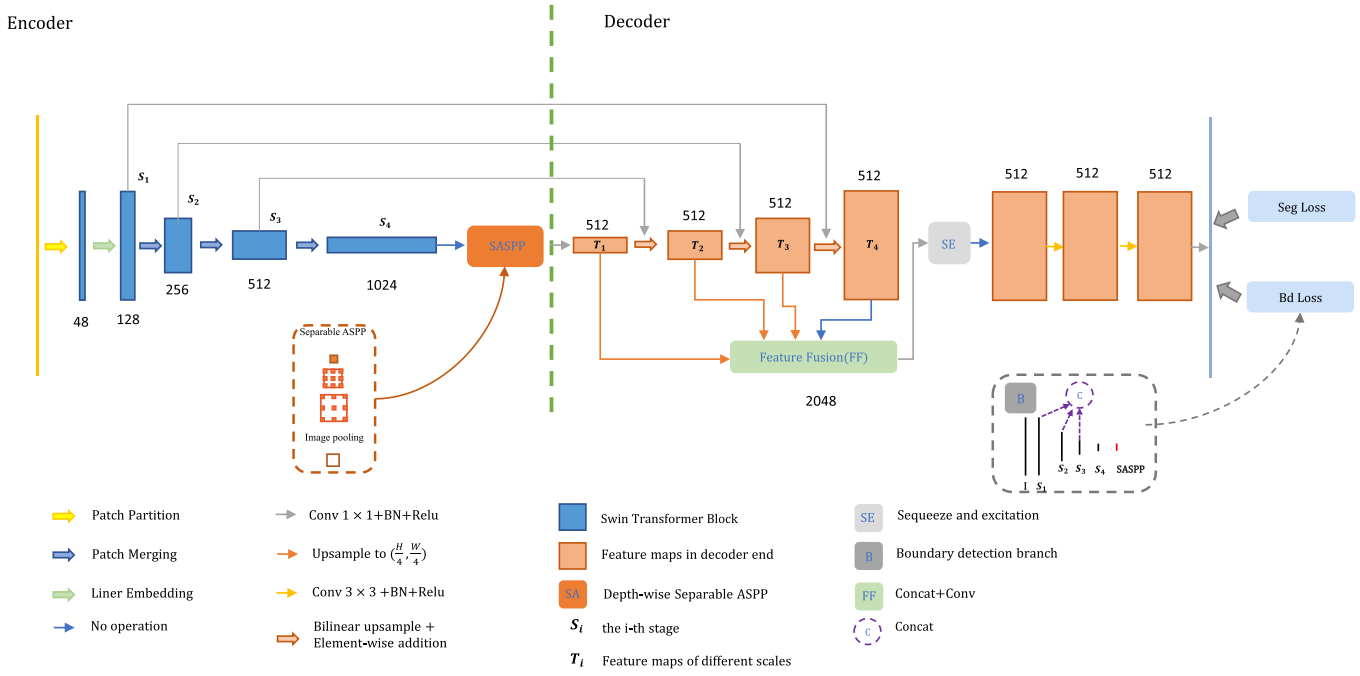


Fig. 1. Pipeline of the proposed transformer and CNN hybrid deep neural network for semantic segmentation of VHR imagery.

on large-scale datasets, it has obtained the state-of-the-art performance. This proves the powerful feature extraction ability of the transformer in image processing and also confirms the effect of inductive bias on model training.

SETR [18] applies the transformer structure to the semantic segmentation task; it uses the transformer structure departed from the ViT as encoder and obtains dense predicted results by applying three different CNN-based decoders. The detection transformer (DETR) [17] constructed the first end-to-end object detection network-based transformer, and it can be easily extended to semantic segmentation task by attaching a decoder head. The model first uses the CNN to extract features from the raw image and then splits the feature maps into a sequence to send them to the transformer, just like the operations in ViT and SETR.

Although proven to have powerful feature extraction capabilities, the existing transformer-based model has a huge number of parameters and increases with the square multiple of the image size, which makes it difficult for transformer-based models to process high-resolution images. In addition, large-scale datasets are required to pretrain the model to ensure the competitive performance of the transformer structure. More importantly, the above transformer structures can only output fixed-size feature maps (such as $(\mathbf{H}/16) \times (\mathbf{W}/16)$), which may lead to insufficient edge positioning accuracy and also limit the expansion of the model in the CNN-style decoder.

To solve the above problems, Xie *et al.* [42] propose a cutting-edge transformer framework for semantic segmentation (SegFormer). It generates hierarchical feature maps through the strategy of overlapped patch merging, replaces the positional encoding in the ViT by mixing 3×3 convolutions and multilayer perceptron (MLP), and constructs a lightweight all-MLP decoder. Liu *et al.* [20] proposes a hierarchical ViT (Swin transformer), which divides the initial image into nonoverlapping patches with a size of 4×4, and the self-attention calculation is limited to a local window composed of a certain number of image patches. Since a single local-window self-attention cannot model the global dependence among pixels, a two-successive transformer structure is further designed by window shifting. This design greatly improves the efficiency of the global self-attention calculation. In addition, the Swin transformer structure generates hierarchical feature maps through concatenation of four adjacent patches, which can be perfectly integrated into the existing encoder–decoder structure.

At present, the application of the transformer in semantic segmentation is still in the preliminary exploration stage. How to build a concise and efficient transformer structure and combine it with the existing CNN model requires further exploration. For a more detailed overview, please refer to [43], [44].

III. Method

In this section, we introduce in detail the semantic segmentation method that combines the Swin transformer encoder and the CNN-based decoder for semantic segmentation of VHR images. Section III-A describes the overall design of our pipeline. Section III-B describes the principle and network structure of the Swin transformer backbone and content about the SASPP block. Finally, Section III-C describes the structure of the CNN-based decoder.

A. Architecture

The pipeline (see Fig. 1) of the proposed transformer and CNN hybrid semantic segmentation network follows an encoder–decoder paradigm and consists of the following modules.

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1) A Swin transformer backbone is applied for feature extraction. Compared with the CNN-based backbone, the Swin transformer does not include inductive bias and thus can better model long-range spatial dependence in the image. Meanwhile, compared with other transformer-based backbones, the Swin transformer has lower computational complexity, faster inference speed, and can output hierarchical feature maps.
2) An SASPP block is inserted in the middle of the network to integrate multiscale contextual information on the basis of the maximum size feature maps.
3) A U-shaped decoder is applied to gradually restore the size of the feature map. Three skip connections are established between encoder and decoder feature maps of the same size to maintain the transmission of local details and enhance the communication of multiscale features. An SE channel attention block is added before segmentation to reallocate weights of the multiple-resolution feature maps for better segmentation results.
4) Using the dual-task design, a boundary detection task is added to assist the semantic segmentation task. Considering the difference between the feature maps output by the Swin transformer and the CNN-based backbones, we designed two boundary detection strategies and combined a more effective one in the proposed network.

B. Swin Transformer-Based Encoder

The encoder consists of two parts: the Swin transformer backbone is used to extract hierarchical feature maps, and the SASPP block is used to capture multiscale contextual information.

1) Swin Transformer Block: The Swin transformer block is the core of the Swin transformer backbone. It consists of two forms of window-based multihead self-attention (W-MSA) block and shift-window-based multihead self-attention (SW-MSA) block. A W-MSA block and an SW-MSA block are connected in sequence to obtain a global spatial dependence in a more efficient way. The detailed calculation process is shown in Fig. 2.

The first Swin transformer block is a W-MSA block; the input features $\mathbf{x}^{l-1}$ flow through the connected layer normalization (LN) and W-MSA layers; and a residual connection is built to obtain the output $\hat{\mathbf{x}}^{l}$. Following that, $\hat{\mathbf{x}}^{l}$ undergoes the connected LN and MLP layers, and the output $\mathbf{x}^{l}$ is finally obtained through another residual connection. The structure of the SW-MSA block is almost the same as the structure of the W-MSA block, except that there is an offset of half the window size in the calculation of the SW-MSA layer. The process can be expressed as follows:

$$\begin{aligned}
\hat{\mathbf{x}}^{l} &= W_{\mathrm{MSA}}\left(\mathrm{LN}\left(\mathbf{x}^{l-1}\right)\right) + \mathbf{x}^{l-1} \\
\mathbf{x}^{l} &= \mathrm{MLP}\left(\mathrm{LN}\left(\hat{\mathbf{x}}^{l}\right)\right) + \mathbf{x}^{l} \\
\hat{\mathbf{x}}^{l+1} &= \mathrm{SW}_{\mathrm{MSA}}\left(\mathrm{LN}\left(\hat{\mathbf{x}}^{l}\right)\right) + \mathbf{x}^{l} \\
\mathbf{x}^{l+1} &= \mathrm{MLP}\left(\mathrm{LN}\left(\hat{\mathbf{x}}^{l+1}\right)\right) + \mathbf{x}^{l+1}. \qquad (1)
\end{aligned}$$

For the calculation of the standard multihead attention layers, please refer to [16], [17] for details. The design of the shifted window is the essence of the Swin transformer backbone. In the following description, we will clearly describe how the Swin transformer backbone models the global spatial dependence through successive W-MSA and SW-MSA blocks based on cross-window connection.

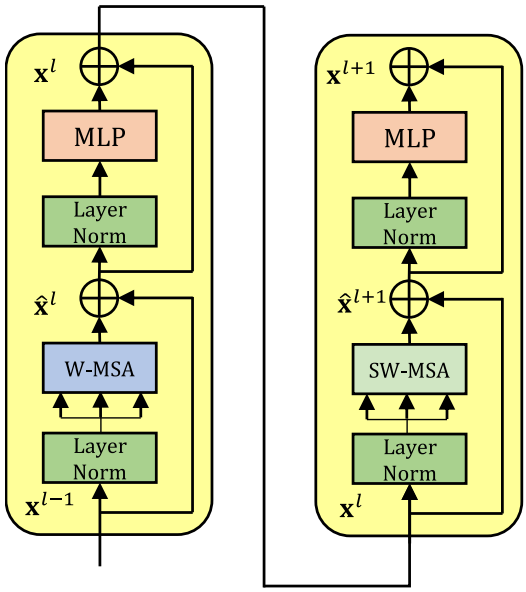


Fig. 2. Two successive Swin transformer blocks. The first is the W-MSA block, and the second is the SW-MSA block.

Taking Fig. 3 as an example, assume that the size of the input feature map is 56 × 56. In the first Swin transformer block, the window division for the W-MSA is shown in Fig. 3(b). Suppose that the window size $\mathbf{W}$ is set to 7; then, the multihead self-attention calculation is limited to 49 pixels in the same red window, and correlation between pixels in different windows are not considered. It is obvious that a single Swin transformer block cannot model the dependencies between global pixels. In the next Swin transformer block, the top 3 × 56 pixel bar and the left 3 × 56 pixel bar are moved to the bottom and right of the image, respectively, as shown in Fig. 3(c), and the window dividing line is moved down and to the right by round down of $(\mathbf{W}/2)$ pixels, respectively, as shown by the green grids in Fig. 3(d). The following SW-MSA is calculated in the new split windows. The design cleverly establishes a cross-window link for the two Swin transformer blocks so that the architecture has the ability to model global spatial dependencies. What is more, compared with directly using all pixels to calculate self-attention, the computational complexity is significantly reduced.

In the actual SW-MSA calculation, there are still some issues that need to be dealt with, such as how to deal with the green window that contains the red and blue areas in Fig. 3(d). The authors' handling is clever; according to their shared code, they masked the invalid self-attention values in local areas (yellow–blue, yellow–red, or blue–red ones), and we followed their processing in our experiments. For a more detailed interpretation, please refer to [20].

2) Swin Transformer Backbone: The architecture of the Swin transformer backbone is simple, consisting of a patch partition module and four stages. The structure of the

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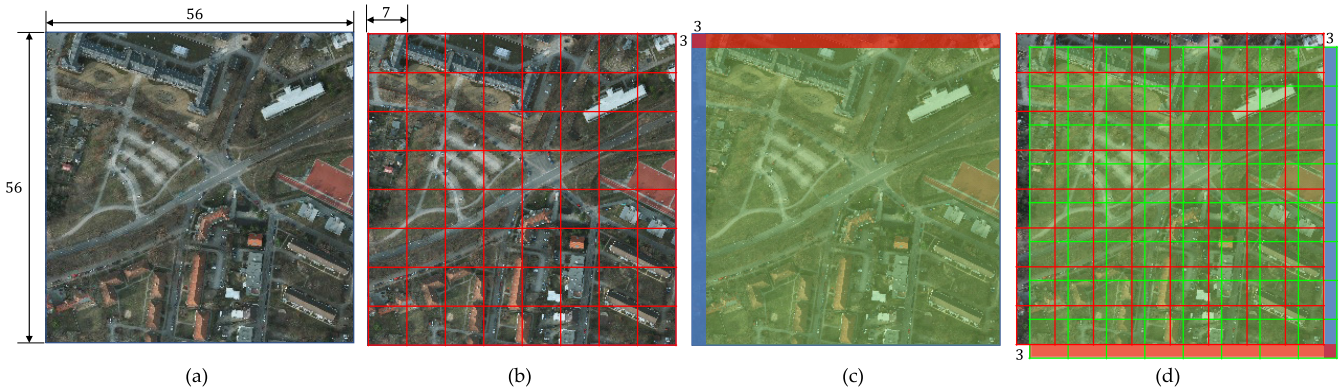


Fig. 3. Cross-window connection built between the W-MSA and SW-MSA blocks for calculation of the global spatial dependence. (a) Input image. (b) Window division for the W-MSA. (c) Shifted pixel bars. (d) Window division for the SW-MSA.

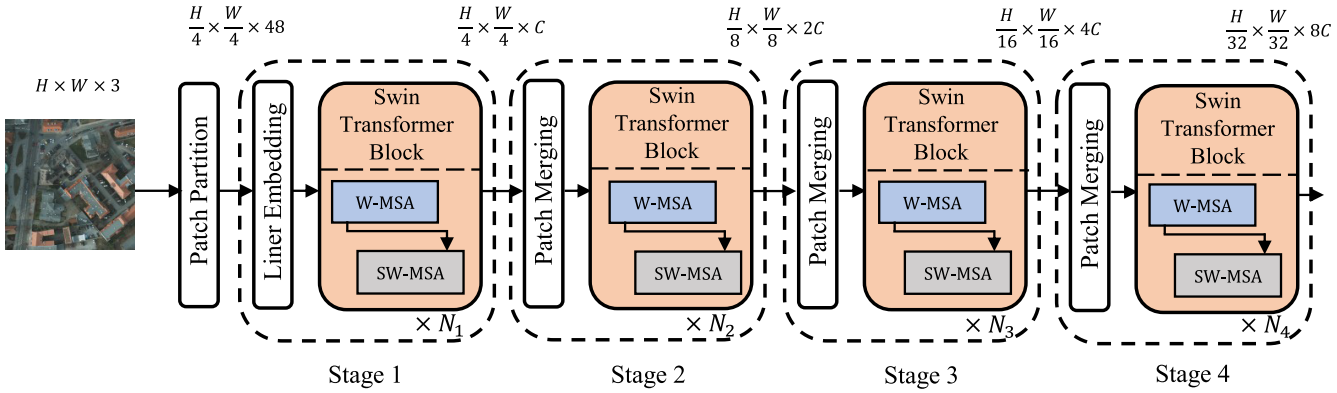


Fig. 4. Architecture of the Swin transformer backbone. The Swin transformer block includes two forms, namely, W-MSA block and SW-MSA block.

four stages is similar; each stage contains a linear embedding/patch merging block and an even number of Swin transformer blocks (see Fig. 4). Specifically, the backbone of the Swin transformer first divides the input image into nonoverlapping patches through patch partition. In the original work of the Swin transformer [20], the size of each patch is 4×4, which means that the image details are better preserved compared with the ViT structure [16]. Then, in the first stage, the patch is linearized and input to consecutive N_1 Swin transformer blocks. Patch merging of Stages 2–4 is slightly different from linear embedding in Stage 1. Four adjacent patches are first concatenated to merge into one. Therefore, the resolution of the feature maps is changed to half, and the channel number is changed four times. Reduce the channel by half through 1×1 convolution, and following that, the features are fed into the subsequent consecutive N_i $(i = 2,3,4)$ Swin transformer blocks.

There are four different versions of the Swin transformer backbones [20]; when dealing with specific tasks, one can select the appropriate version of the Swin transformer backbone according to the actual performance of the model and the limitations of the hardware conditions. In our experiment, two versions of the Swin-small (SwinS) and Swin-base (SwinB) backbones are used, and their detailed architectures are shown in Table I.

3) Multiscale Context Extraction: We use the classic SASPP [45] block to extract multiscale features. Features of different receptive fields are extracted from the feature maps output by Stage 4. The SASPP block contains five parallel branches, which are: 1) a global average pooling layer; 2) three 3×3 depthwise separable atrous convolutions with the rate of {2,3,4}; and 3) a 1×1 convolution. Rates are set according to the size of the feature maps in Stage 4. We crop the original image to 300×300 as inputs. Therefore, the output feature of Stage 4 is 10×10. In this condition, the rates of the atrous convolutions are set to {2,3,4}, which corresponds to a distance of {5,7,9} on the feature maps of Stage 4 and a distance of {160,224,288} on the original image.

C. CNN-Based Decoder

We apply a U-shaped decoder to gradually recover the size of the feature map and predict the semantic segmentation result. Three skip connections are built between the encoder and decoder feature maps of the same size to maintain the transmission of local details and enhance the communication of multiscale features. In addition, an SE block is added before segmentation to selectively enhance the weight of more important features and reduce the weight of less concerned features. The architecture of the CNN-based decoder is shown in Fig. 5.

First, the channel dimension of tensors output by Stages 1–3 is converted to 512 through 1×1 convolution to prepare for feature fusion. The form of the feature maps output by the SASPP block can be marked as $\mathbf{T}_1 \in \mathbb{R}^{(\mathbf{H}/32)\times(\mathbf{W}/32)\times 512}$.

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TABLE I

DETAILED ARCHITECTURE OF SWINS AND SWINB. THE SWIN TRANSFORMER BLOCKS ARE IN SQUARE BRACKETS, "HEAD" REPRESENTS THE NUMBER OF HEADS IN EACH SWIN TRANSFORMER BLOCK, AND THE NUMBER OF STACKED SWIN TRANSFORMER BLOCKS AT EACH STAGE IS DISPLAYED OUTSIDE THE BRACKETS. IN THIS CASE, THE SIZE OF THE INPUT IMAGE IS 300 × 300

	Downsp.rate (output size)	SwinS	SwinB
Stage 1	$4 \times (75 \times 75)$	concat 4×4, 96-d, LN $\begin{bmatrix} win.sz.7 \times 7, \\ dim\ 96, head\ 3 \end{bmatrix} \times 2$	concat 4×4, 128-d, LN $\begin{bmatrix} win.sz.7 \times 7, \\ dim\ 128, head\ 4 \end{bmatrix} \times 2$
Stage 2	$8 \times (38 \times 38)$	concat 2×2, 192-d, LN $\begin{bmatrix} win.sz.7 \times 7, \\ dim\ 192, head\ 6 \end{bmatrix} \times 2$	concat 2×2, 256-d, LN $\begin{bmatrix} win.sz.7 \times 7, \\ dim\ 256, head\ 6 \end{bmatrix} \times 2$
Stage 3	$16 \times (19 \times 19)$	concat 2×2, 384-d, LN $\begin{bmatrix} win.sz.7 \times 7, \\ dim\ 384, head\ 12 \end{bmatrix} \times 2$	concat 2×2, 512-d, LN $\begin{bmatrix} win.sz.7 \times 7, \\ dim\ 512, head\ 6 \end{bmatrix} \times 2$
Stage 4	$32 \times (10 \times 10)$	concat 2×2, 768-d, LN $\begin{bmatrix} win.sz.7 \times 7, \\ dim\ 768, head\ 24 \end{bmatrix} \times 2$	concat 2×2, 1024-d, LN $\begin{bmatrix} win.sz.7 \times 7, \\ dim\ 1024, head\ 6 \end{bmatrix} \times 2$

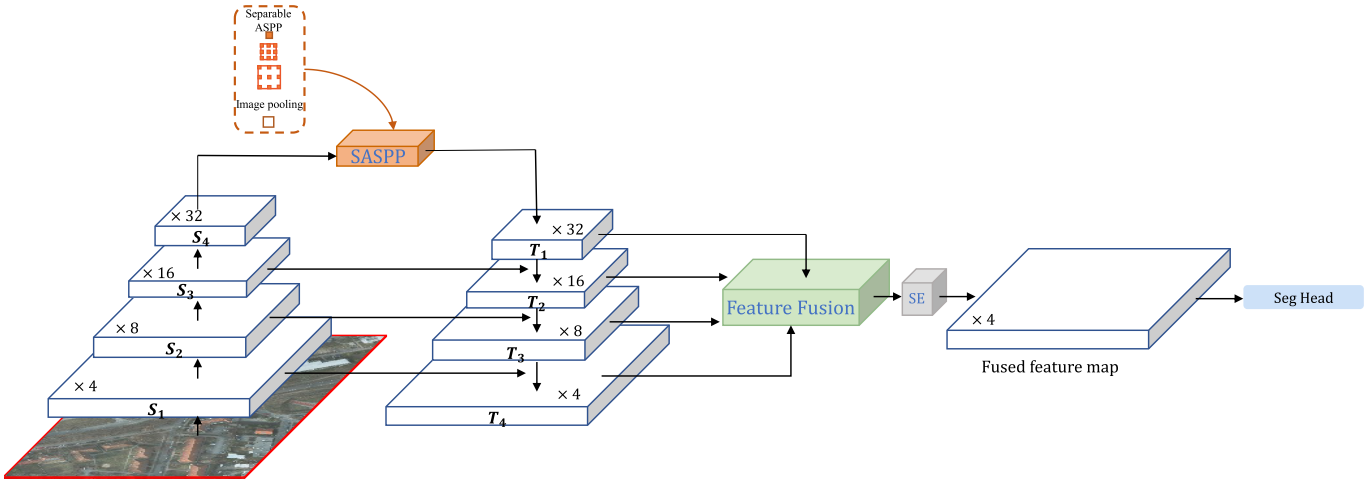


Fig. 5. Structure of the CNN-based decoder module.

It connects two branches: one branch links to the feature fusion block, and the other branch is first upsampled to $[(\mathbf{H}/16), (\mathbf{W}/16)]$ and then merged with the feature maps output by Stage 3 in a way of elementwise addition. The output feature maps can be marked as $\mathbf{T}_2 \in \mathbb{R}^{(\mathbf{H}/16)\times(\mathbf{W}/16)\times 512}$. The $\mathbf{T}_2$ also links to two branches: one branch is the fusion module, and the other branch continues to be upsampled to $[(\mathbf{H}/8), (\mathbf{W}/8)]$ and then fused with the feature maps output by Stage 2 through elementwise addition. The result can be marked as $\mathbf{T}_3 \in \mathbb{R}^{(\mathbf{H}/8)\times(\mathbf{W}/8)\times 512}$. The $\mathbf{T}_3$ still connects two branches: one branch is the feature fusion block, and the other branch continues to be upsampled and fused with the feature maps output by Stage 1 to obtain $\mathbf{T}_4 \in \mathbb{R}^{(\mathbf{H}/4)\times(\mathbf{W}/4)\times 512}$. $\mathbf{T}_4$ is finally input to the feature fusion module.

In the feature fusion module, $\mathbf{T}_1$–$\mathbf{T}_3$ are first upsampled to the same resolution as $\mathbf{T}_4$ through bilinear interpolation, and then, the feature maps are concatenated to construct $\mathbf{T}_0 \in \mathbb{R}^{(\mathbf{H}/4)\times(\mathbf{W}/4)\times 2048}$. In order to enhance the interaction between the channels, we use the SE block to recalibrate the weights of the feature maps. The purpose is to selectively excite features that are more effective for semantic segmentation from features of multiple scales. The feature maps output by the SE block then undergo two consecutive 3×3 convolutions and one 1×1 convolution to obtain the final predicted semantic segmentation result.

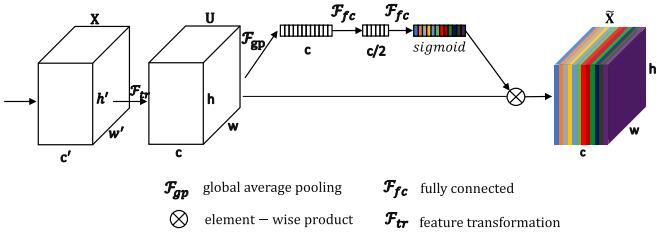


Fig. 6. Structure of the SE block.

The paradigm composed of the Swin transformer backbone and the aforementioned CNN-based decoder is the baseline of our work. According to the specific use of SwinB or SwinS as the backbone, the models can be denoted as SwinB-CNN or SwinS-CNN, respectively.

1) SE Block: Assuming that the input tensor $\mathbf{X} \in \mathbb{R}^{h'\times w'\times c'}$ is transformed to tensor $\mathbf{U} \in \mathbb{R}^{h\times w\times c}$ by transformation $\mathcal{F}_{\text{tr}}$, we will take the implementation of the SE operation on $\mathbf{U} \in \mathbb{R}^{h\times w\times c}$ as an example to illustrate the construction of the SE block.

As shown in Fig. 6, the spatial dimension of $\mathbf{U} \in \mathbb{R}^{h\times w\times c}$ is first squeezed by global average pooling to generate a vector $\mathbf{P} \in \mathbb{R}^c$. Therefore, each element in the vector can be regarded as embedding the global information of the corresponding channel. Specifically, the $\mathbf{c}$th element in the vector $\mathbf{P} \in \mathbb{R}^c$

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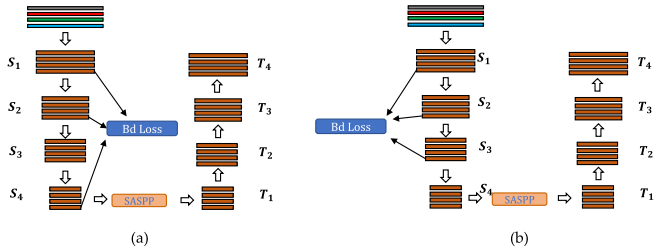


Fig. 7. Two different boundary detection strategies applied in this work.

can be calculated according to the following formula:

$$\mathbf{P_c} = \mathcal{F}_{\text{gp}}(\mathbf{U}_c) = \frac{1}{\mathbf{H} \times \mathbf{W}} \sum_{i=1}^{\mathbf{H}} \sum_{j=1}^{\mathbf{W}} \mathbf{U}_c(i, j). \tag{2}$$

Subsequently, the weight vector $\mathbf{V} \in \mathbf{R}^c$ is obtained through two consecutive fully connected layers, and its detailed structure can be expressed as

$$\mathbf{V} = \delta_{\mathbf{2}}(\mathcal{F}_{\text{fc2}}(\delta_{\mathbf{1}}(\mathcal{F}_{\text{fc1}}(\mathbf{P})))) \tag{3}$$

where $\mathcal{F}_{\text{fc1}}$ and $\mathcal{F}_{\text{fc2}}$ represent two fully connected layers, and $\delta_{\mathbf{1}}$ and $\delta_{\mathbf{2}}$ represent ReLU and sigmoid activations, respectively. Numbers of neurons in $\mathcal{F}_{\text{fc1}}$ and $\mathcal{F}_{\text{fc2}}$ are set to $(\mathbf{c}/2)$ and $\mathbf{c}$, respectively. Finally, elementwise product is performed between the weight vector $\mathbf{V}$ and the input feature $\mathbf{U}$ to obtain the final output tensor $\widetilde{\mathbf{X}}$.

2) Boundary Detection: Based on the exploration of studies [7], [31], [33], combining edge constraints in the CNN-based remote sensing image segmentation network can effectively improve the performance of the model. Therefore, we adopt a dual-task network design and use the auxiliary boundary detection task to improve the accuracy of boundary pixel positioning and reduce the salt and pepper noise for semantic segmentation.

In order to avoid the introduction of many new parameters, the boundary detection task and the semantic segmentation task share the same backbone. Feature maps in the encoder–decoder structure are directly used to predict the edge pixels. We tried two different boundary detection structures in this work, as displayed in Fig. 7. The first structure applies the feature maps output by Stages 1, 2, and 4 to predict the edge pixels, and the second structure applies the feature maps output by Stages 1–3 to predict the edge pixels. The first design draws on the experience from CNN-based edge extraction methods, using shallow features to preserve local details, and using deep features to suppress complex texture [7], [46] [36]. The second design is the structure that we adjusted based on the first design according to the characteristics of the Swin transformer backbone. The reason of our adjustment and the comparative experiments of the two strategies are displayed in Sections V-A and V-D, respectively.

The most recent research boundary metric (BF1) is applied to evaluate the accuracy of boundary detection [47], and the $1 - \text{BF1}$ is used as the loss function for model training. $\text{BF1} = 2 \times (\text{precision} \times \text{recall})/(\text{precision} + \text{recall})$, which can comprehensively evaluate the recall and precision of the boundary detection results. For details about the differentiable surrogate used for training and the calculation steps in computer language, please refer to [7], [47].

We use the most widely used cross-entropy loss to evaluate the accuracy of semantic segmentation. The total loss is the weighted sum of the edge loss and the segmentation loss

$$\text{loss}_{\text{total}} = \alpha \cdot \text{loss}_{\text{seg}} + \beta \cdot \text{loss}_{\text{bd}} \tag{4}$$

where α and β are the weights of semantic segmentation and boundary detection, respectively. Theoretically, the setting of α and β should be dynamic. Because the edge positioning in the early stage of model training is not accurate, the effect of boundary detection on semantic segmentation in the later stage of model training also needs to be analyzed through multiple sets of experiments. However, according to our experiments on multiple groups of α and β combinations using the pretrained model, the values of α and β have little effect on the convergence speed and the accuracy of the model. Therefore, in our dual-task experiments, we uniformly set the weight of α to 1 and β to 0.15.

IV. Experimental Setup

A. Datasets

We evaluate the proposed model on two public datasets provided by the International Society for Photogrammetry and Remote Sensing (ISPRS) for 2-D semantic labeling. They both contain three band optical images and the corresponding nDSM. In addition to full reference ground truth, the organizer of the contest also provided edge-eroded reference to reduce the impact of uncertain border definitions on the evaluation.

1) Potsdam Benchmark: Potsdam is a typical historic city in northeastern Germany. It has large buildings, narrow streets, and dense traffic. The ground sampling distance of the Potsdam dataset is 5 cm. It contains 38 image blocks; among them, 24 blocks are used for training, and the remaining 14 blocks are used for testing. The size of each image block is 6000 × 6000 pixels. Six classes of objects, namely, building (blue), car (yellow), low vegetation (cyan), impervious surface (white), tree (green), and background (red), are labeled. The background class includes mainly nonclassifiable atypical targets.

2) Vaihingen Benchmark: Vaihingen is a small town in southern Germany. The main building types in Vaihingen are adjacent buildings and small multistory buildings with less traffic in the town. The ground sampling distance of the Vaihingen dataset is 9 cm. It contains 33 image blocks, of which 17 blocks are used for training, and the remaining 16 blocks are used for testing. The size of each image block is different, about 2000 × 2500 pixels; the required labeling classes are the same as the Potsdam dataset.

B. Evaluation Metrics

For both datasets, we calculate the confusion matrices and extract the overall accuracy (OA) and the F1 score of each class to evaluate the semantic segmentation results. The F1 score is a comprehensive evaluation of precision and recall of a specific class, and the calculation

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formula is given as follows:

$$\text{precision} = \frac{\mathbf{TP}}{\mathbf{TP} + \mathbf{FP}} \tag{5}$$

$$\text{recall} = \frac{\mathbf{TP}}{\mathbf{TP} + \mathbf{FN}} \tag{6}$$

$$\mathbf{F}_1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{7}$$

where **TP**, **FP**, and **FN** stand for true positive, false positive, and false negative, respectively. In calculation, **TP** is the number of pixels where predicted condition and actual condition are both true, and **FP** is the number of pixels where predicted condition is true and the actual condition is false, while **FN** is the number of pixels where predicted condition is false and the actual condition is true. The OA is the ratio of the number of correctly predicted pixels to the total number of pixels.

The organizer of the contest provided the ground truth of full reference and edge-eroded reference. The edge-eroded reference uses a disk of a three-pixel radius to erode the boundaries between objects, which can reduce the impact of uncertain border definitions on the evaluation. Therefore, the precision value evaluated by the full reference is a bit lower than that of the edge-eroded reference. Since most recent studies based on the ISPRS datasets use the edge-eroded reference, we applied the edge-eroded reference for accuracy evaluation in all of our experiments to maintain the consistency of the article. Therefore, one can easily compare our work with other studies.

C. Implementation Details

In both training and testing sets, raw images are split into patches with a size of 600 × 600 pixels, and the overlap between two adjacent patches is 300 pixels. In the data prepossessing stage, we use a series of standard data augmentation strategies. Random resize (ratio 0.5∼2), random crop, random flip, adding photometric distortion, and band normalization operations are applied to the input images in sequence. Increasing the size of the input image can achieve higher model accuracy, as we discussed in Section V-B. However, in order to balance the semantic segmentation results and training time consumption, the input image sizes are all set to 300 × 300 in our work, except for experiments in Section V-B.

We train all of our transformer-based models use adaptive moment estimation (AdamW) optimizer [48]. The initial learning rate is set to 6×10^{-5}, and the weight decay is set to 0.01. A polynomial learning rate policy is applied with the formula: $\text{lr} = \text{lr}_0 * (1 - i/\mathbf{T}_i)^{\text{power}}$. Here, lr_0 represents the initial learning rate, i represents the current number of iterations, $\mathbf{T}_i$ represents the total number of iterations, and the "power" is set to 0.9. In the initial training stage, we applied a warm-up of 1500 iterations to slow the model's early overfitting of minibatch. All experiments in this article are initialized with pretrained weights to speed up the convergence of the model.

Models are trained on a machine equipped with a GPU of NVIDIA 1080ti with 11-GB memory. Taking into account the limitations of hardware conditions, our batch size is set to 2, and the number of max iterations for all training is set to 160k.

TABLE II

COMPARISON OF OA (%), PARAMETER SIZE (MILLION), AND COMPUTATION COMPLEXITY (GFLOPS) BETWEEN SWIN TRANSFORMER AND OTHER STATE-OF-THE-ART BACKBONES BASED ON POTSDAM BENCHMARK. THE INPUT IMAGE SIZE IS SET TO 300 × 300, AND THE RESULTS ARE EVALUATED USING AN EDGE-ERODED REFERENCE

BackBone	Decoder	OA	#param	GFLOPs
ResNeSt-101 [22]	deeplabv3+	89.32	64 M	130
ResNet-101 [13]	deeplabv3+	88.84	60 M	179
HRNet-W48 [14]	-	88.90	66 M	66
ViT [16]	SETR-PUP	77.03	308 M	-
SegFormer-B4 [42]	All-MLP	84.51	64 M	48
SwinB [20]	deeplabv3+	89.58	104 M	114
SwinS [20]	deeplabv3+	88.36	66 M	88

V. EXPERIMENTS AND RESULTS

In this section, we first conduct a series of experimental analyses on the proposed architecture and then compare the proposed architecture with the state-of-the-art models in the remote sensing community. The ISPRS datasets provide both optical images and nDSM data for model training. Since nDSM data contain height information, it can effectively improve the performance of the model. However, in most cases, nDSM data are unavailable or not provided. Considering the versatility of the model, nDSM was not used in the series of experimental analyses based on our proposed architecture, while, in comparison with other state-of-the-art algorithms, we stacked the nDSM and the optical images as input channels to pursue more competitive semantic segmentation accuracy. Please note that almost all advanced methods in published papers used the nDSM data.

A. Comparison of Swin Transformer With Other State-of-the-Art Backbones

We first compared the feature extraction capabilities of the Swin transformer backbone with some of the most popular backbones on the Potsdam dataset. Except for HRNet [14], ViT [16], and SegFormer [42], all backbones use a deeplabv3+ decoder. The HRNet maintains the high resolution of the input image; its low-resolution feature map can be directly upsampled to the same size as the high-resolution feature map. Multiscale features are concatenated to predict the semantic segmentation results. The ViT backbone can only output feature maps at a (1/16) size of the input imagery, and we applied the progressive upsampling (PUP) decoder as it is recommended in the SETR [18]. The SegFormer applies a lightweight decoder that consists of multiple MLP layers for efficient segmentation. All models' parameters are initialized with pretrained weights. The max number of iterations is set to 160 000, the input image size is set to 300 × 300, and the semantic segmentation results are evaluated based on the full reference ground truth.

As shown in Table II, SwinB achieved the highest OA of 89.58%, slightly higher than the best CNN-based backbone ResNeSt. Compared with ViT, the number of parameters of SwinB has dropped by about two-thirds to 104M, but still higher than the CNN-based backbones. ViT does not perform well for a large number of parameters and low OA. Model sizes of SwinS and SegFormer are basically the same, SwinS

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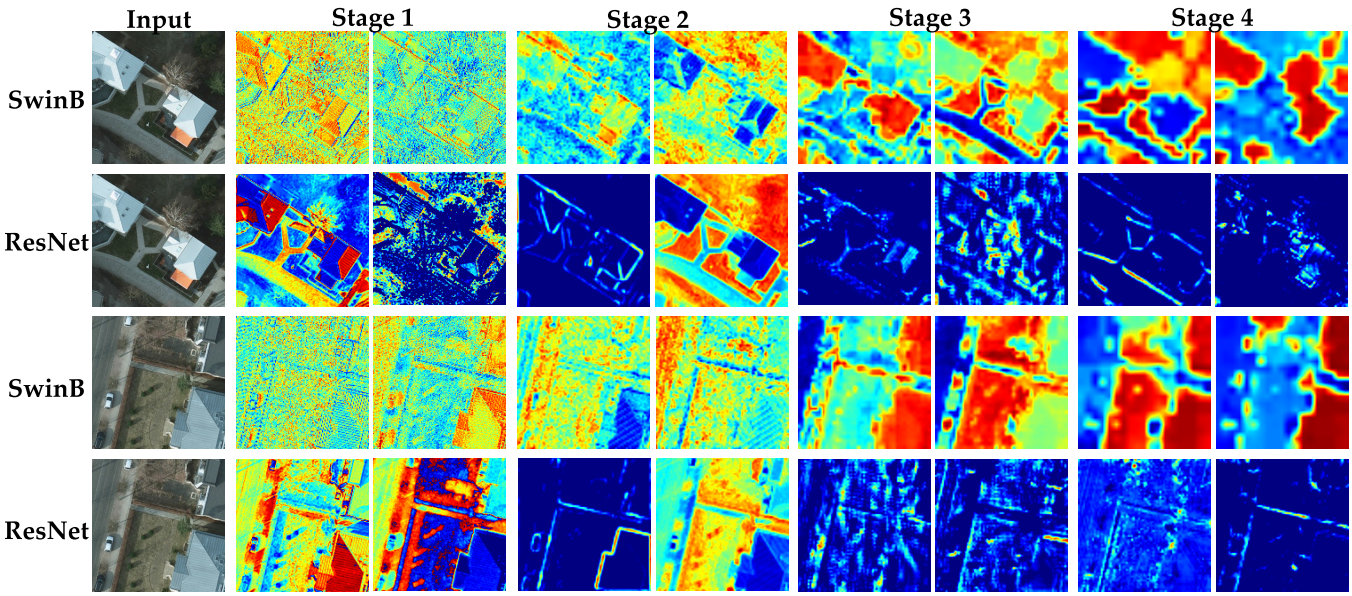


Fig. 8. Examples of feature maps extracted by SwinB and ResNet-101. The leftmost column shows the input RGB images. Since the numbers of the output features of each stage are 128, 256, 512, and 1024 for both backbones, we randomly select two output channels of each stage to display for comparison. They are displayed in the columns from left to right and from the first stage to the fourth stage. Feature maps are rendered using COLORMAP_JET pseudocolors for better visual discrimination, with red representing higher pixel values and blue representing lower pixel values in the feature maps.

has higher accuracy, and SegFormer has lower computation complexity. ResNeSt has a good tradeoff between the size and the accuracy of the model, but the model's computation complexity is high.

Existing studies have proved that SETR's semantic segmentation accuracy can be slightly higher than CNN-based models when pretrained on a large dataset; however, we have not obtained similar good results. A similar situation also appears on the SegFormer backbone; there is a small gap between its accuracy and the accuracy of the popular CNN-based backbones or the Swin transformer backbone. Considering that the amount of data in the Potsdam dataset may be insufficient, we have also tested their performance on a larger aerial imagery for roof segmentation (AIRS) dataset [49] for reference (Appendix A), and the results are similar to those on the Potsdam dataset. Our results provide a baseline of these popular backbones on the Potsdam dataset when the batch size is limited to 2.

Transformer- and CNN-based backbones are derived from completely different feature extraction principles. To understand the characteristics of the feature maps extracted by the two, we visualized their respective feature maps for comparative analysis. This is also important for reasonable network design in the decoder module. Some examples are shown in Fig. 8.

The figures show that the shallow layers of the CNN backbone mainly extract basic image descriptors, such as points, lines, and local textures, while the deep layers extract high-level semantic information, and the visual effect appears "abstract." For the Swin transformer backbone, the shallow feature maps show "abstract," while the deep feature maps show obvious "clustering" characteristics. This is mainly because CNN extracts feature through convolution kernels, while transformer extracts feature through self-attention. In self-attention, the weight of a specific pixel is calculated by the sum of its similarity with each pixel in the image. In the shallow layer, the patch is small, so the pixel's "receptive field" is small, and the similarity calculation between pixels may not be stable enough. In the deep layer, with the continuous merging of adjacent patches, the "receptive field" of each pixel continues to increase, so the calculation of the similarity between pixels is more stable.

It should be noted that the effective use of feature maps is closely related to the accuracy of the vision task. In CNN-based boundary detection, the common paradigm is to use shallow features to preserve local details and use deep features to suppress complex textures [7], [24]. Therefore, in our experiment, we first use the feature maps output by Stages 1, 2, and 4 to obtain the edge pixels. However, the results are not as good as the CNN-based model. We found possible explanations for this phenomenon from the visualization of feature maps: 1) the shallow feature map of the transformer-based model is not good at extracting local image descriptions as the CNN-based model and 2) the feature map output by Stage 3 of the transformer-based model already has good clustering results, and the resolution is better than Stage 4. Therefore, feature maps in Stage 3 may be more suitable for boundary detection than Stage 4. In response to these assumptions, we designed more detailed comparative experiments and analysis in Section V-D.

B. Influence of Input Resolution

In the original work of the Swin transformer [20], it is proven that the input image size has an influence on the performance of the model. We have completed similar comparative experiments on the Potsdam benchmark based on the SwinB backbone. In this set of experiments, we applied the

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TABLE III

INFLUENCE OF THE INPUT SIZE OF THE IMAGE ON THE F1 SCORE (%) AND OA (%) OF SEMANTIC SEGMENTATION. EVALUATIONS ARE BASED ON THE "SWINB + 3SKIPS" ARCHITECTURE USING THE POTSDAM DATASET WITH AN EDGE-ERODED REFERENCE

Input size	Building	Car	Low_veg	Imp.	Tree	OA	time
224 × 224	95.6	95.1	87.3	92.9	85.6	89.7	**11h**
300 × 300	95.9	95.3	87.6	93.1	86.1	90.0	14h
448 × 448	96.2	96.0	87.7	93.2	86.4	**90.3**	22h

U-shaped decoder with all three skip connections (SwinB + 3skips), while the SE block and the boundary detection task are not introduced. The default input resolution of the proposed "SwinB + 3skips" is 300 × 300. We train the model on a smaller input image size of 224 × 224 and on a larger input image size of 448 × 448 additionally.

As indicated in [20] and [50], for the "SwinB + 3skips" semantic segmentation structure proposed in this article, increasing the resolution of the image will help improve the OA and the F1 score of each class but will also bring about a significant increase in training time (see Table III). Therefore, a suitable input image size can be selected according to the specific task requirements and the hardware limitations.

C. Architecture Ablation Study

In this section, we design comprehensive ablation experiments to analyze the effectiveness of each component in our proposed decoder module.

1) Effectiveness of Skip Connection in the Decoder Module: Taking inspiration from the U-Net series, we established multiple skip connections between the encoder and decoder paths to maintain the transmission of local details and enhance the communication of multiscale features. In this section, we design several experiments to test the effectiveness of the skip connections in our model. Experiments are accomplished based on the SwinB backbone on the Potsdam benchmark, and the input image size is set to 300 × 300. In order to avoid introducing additional interference, we did not add the boundary detection task or the SE block to assist the semantic segmentation in this set of experiments.

As shown in Fig. 9, the proposed model applies a SwinB encoder and a decoder with three skip connections (SwinB + 3skips). The four sets of comparative experiments are given as follows.

1) Retain only one skip connection of $S_1 \rightarrow T_4$. T_3 is upsampled and elementwise addition with S_1 to obtain the T_4.
2) Retain only one skip connection of $S_2 \rightarrow T_3$. T_2 is upsampled and elementwise addition with S_2 to obtain the T_3.
3) Retain only one skip connection of $S_3 \rightarrow T_2$. T_1 is upsampled and elementwise addition with S_3 to obtain the T_2.
4) Do not use any skip connection in the encoder–decoder framework. The resolution of feature maps is retained by three times 2 × upsampling of T_1.

It is to be known that, after each upsampling, two consecutive convolutions with a kernel size of 3 × 3 are applied. The results of the quantitative evaluation are shown in Table IV.

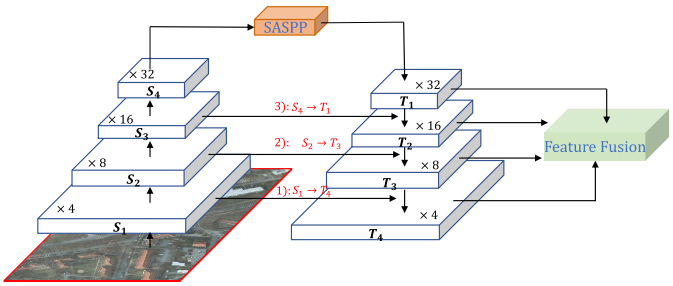


Fig. 9. Three different skip connections built in the encoder–decoder structure.

TABLE IV

PERFORMANCE EVALUATION (F1 SCORE AND OA, %) OF NETWORKS WITH ADDING DIFFERENT NUMBERS AND STYLES OF SKIP CONNECTIONS IN THE ENCODER–DECODER ARCHITECTURE. THE INPUT IMAGE SIZE IS SET TO 300 × 300. RESULTS ARE BASED ON THE POTSDAM DATASET USING AN EDGE-ERODED STATISTICS. THE HIGHEST ACCURACY IS MARKED IN BOLD, AND THE SECOND HIGHEST ACCURACY IS MARKED IN UNDERLINE

	Building	Car	Low_veg	Imp.	Tree	OA
No skip	95.1	92.4	84.7	91.6	84.9	88.4
1 skip: $S_1 \rightarrow T_4$	95.1	**96.0**	87.2	92.7	<u>85.9</u>	<u>89.5</u>
1 skip: $S_2 \rightarrow T_3$	95.2	94.8	<u>87.3</u>	<u>92.8</u>	85.5	89.2
1 skip: $S_3 \rightarrow T_2$	<u>95.6</u>	92.6	87.2	92.6	85.2	89.1
3 skips (SwinB + 3skips)	**95.9**	<u>95.3</u>	**87.6**	**93.1**	**86.1**	**90.0**

The proposed decoder with three skip connections achieves the highest OA of 90.0%, which is 1.6 percentage points higher than the model without any skip connections. Adding any skip will improve the OA of the model. Among them, adding a single $S_1 \rightarrow T_4$ skip connection improves the accuracy of the model the most, which is 1.1 percentage points; adding a single $S_2 \rightarrow T_3$ or $S_3 \rightarrow T_2$ skip connection increases the accuracy of the model by 0.8 and 0.7 percentage points, respectively.

From the details of each class, the model with all three skip connections achieved the highest F1 score in four classes of building, low vegetation, impervious, and tree, while the model with only one $S_1 \rightarrow T_4$ skip connection achieved the best performance in the car class. In addition, adding one $S_2 \rightarrow T_3$ connection is more effective for impervious surface class, while adding one $S_3 \rightarrow T_2$ skip connection is more effective for building class. The results prove that different connections convey different kinds of effective information.

2) Effect of SE Block in the Decoder Module: The design of the SE block can filter the features that are most relevant to the semantic segmentation results and increase their weights for feature augmentation. In order to show the effect of the SE block in our model, we compared the semantic segmentation results when the SE block was added or not. Experiments are accomplished on the Potsdam benchmark based on the backbone of SwinB and SwinS; the decoder module includes all three skip connections; and the input image size is set to 300 × 300. In order not to introduce new influential factors, no boundary detection task is added to the network.

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TABLE V

QUANTITATIVE ANALYSIS (F1 SCORE AND OA, %) OF THE EFFECT OF THE SE BLOCK IN THE PROPOSED MODEL BASED ON THE POTSDAM DATASET. THE INPUT IMAGE SIZE IS SET TO 300 × 300, AND THE RESULTS ARE BASED ON AN EDGE-ERODED REFERENCE STATISTICS. CLASS WITH A RELATIVELY LARGER INCREASE IN F1 SCORE IS MARKED IN **BOLD**

	Building	Car	Low_veg	Imp.	Tree	OA
SwinB+3skips	95.9	**95.3**	87.6	93.1	**86.1**	90.0
SwinB+3skips+SE	96.0	**95.9**	87.7	93.2	**86.9**	90.3
SwinS+3skips	95.2	**95.1**	87.0	92.4	**84.6**	88.6
SwinS+3skips+SE	95.6	**95.6**	87.2	92.2	**85.8**	89.0

TABLE VI

MEAN AND STANDARD DEVIATION OF THE OA OF FOUR METHODS FOR THREE TIMES MODEL TRAINING

	SwinB+3skips	SwinB+3skips+SE	SwinS+3skips	SwinS+3skips+SE
mean	89.97	90.31	88.60	88.99
stddev	0.052	0.044	0.058	0.050

a) Quantitative analysis: As shown in Table V, based on the SwinB backbone, the baseline "SwinB + 3skips" reached an OA of 90.0%, and the addition of the SE block increases the OA of the model by 0.3 percentage points. From a detailed point of view, the SE block is more effective for the class of car and tree, and the F1 scores of car and tree have increased by 0.6 and 0.8 percentage points, respectively, while the F1 scores of building, low vegetation, and impervious is not significantly improved. Based on the SwinS backbone, the baseline "SwinS + 3skips" reached an OA of 88.6%, and the addition of the SE block increases the OA of the model by 0.4 percentage points. From a detailed point of view, significant F1 score improvements are obtained in the two classes of car and tree.

The strategy of adding SE block does not greatly improve the OA of the model, and the improvement that it brings is about 0.3–0.4. Therefore, we count the mean and standard deviation of each model for three repetitions of training. As shown in Table VI, the performance of multiple models on the Potsdam dataset is stable, with a standard deviation of approximately 0.05%. This shows that the effectiveness of the SE block on SwinB and SwinS is consistent; hence, we include the SE block in our encoder–decoder framework.

b) Visualization analysis: Looking back at Fig. 5, the decoder module outputs feature maps of four different scales, namely, T_1–T_4. The small-size feature maps are upsampled to the same size as T_4 for feature fusion. The SE block reallocates the weights of the multiresolution feature maps for feature augmentation. Theoretically, the channel weights calculated by the SE block represent their relevance to the semantic segmentation results. As for the sigmoid operation, the weight of each channel is in the range of 0–1. Due to too many channels (2048 in total and 512 in each scale), the scatter plot cannot clearly show the weight distribution. Therefore, we calculate the sum of the weights of all channels in each scale for statistics. The experiment is performed based on both the SwinB-CNN and SwinS-CNN models on the Potsdam benchmark. The statistical results are shown in Fig. 10, and we also provide the initial scatter plots in Appendix B for additional reference.

TABLE VII

PERFORMANCE COMPARISON (F1 SCORE AND OA, %) OF DIFFERENT BOUNDARY DETECTION STRATEGIES BASED ON THE SWINB-CNN FRAMEWORK ON THE POTSDAM DATASET. THE INPUT IMAGE SIZE IS SET TO 300 × 300. RESULTS ARE BASED ON AN EDGE-ERODED REFERENCE STATISTIC

	Building	Car	Low_veg	Imp.	Tree	OA
SwinB-CNN	96.0	95.9	87.7	93.2	86.9	90.3
SwinB-CNN+BD$_{1,2,4}$	96.5	96.1	87.8	93.4	87.4	90.7
SwinB-CNN+BD$_{1,2,3}$	**96.7**	**96.3**	**88.0**	**93.6**	**88.0**	**91.0**

TABLE VIII

PERFORMANCE COMPARISON (F1 SCORE AND OA, %) OF DIFFERENT BOUNDARY DETECTION STRATEGIES BASED ON THE SWINB-CNN FRAMEWORK ON THE VAIHINGEN DATASET. THE INPUT IMAGE SIZE IS SET TO 300 × 300. RESULTS ARE BASED ON AN EDGE-ERODED REFERENCE STATISTIC

	Building	Car	Low_veg	Imp.	Tree	OA
SwinB-CNN	94.8	86.1	83.3	91.6	89.2	89.8
SwinB-CNN+BD$_{1,2,4}$	95.1	86.4	83.4	92.0	89.4	90.1
SwinB-CNN+BD$_{1,2,3}$	**95.3**	**86.9**	**83.6**	**92.2**	**89.6**	**90.4**

According to Fig. 10(a), we can see that, in the SwinB-CNN framework, the feature maps of T_4 are obviously given the highest weights, while the feature maps of T_1 are given the lowest weights. In addition, T_2 that combined the features maps of S_3 is given the second highest weights. This is consistent with our conjectures derived from the visual analysis of the feature map; compared with S_4, S_3 has good enough clustering ability, and its resolution is higher. The distribution of weights in SwinS-CNN seems to be somewhat different; the feature maps in T_3 with the second highest resolution get the highest weight allocation. This is presumably because the numbers of input channels and "heads" in Stage 1 of SwinS are both less than that of SwinB (C = 96 versus C = 128; head = 3 versus head = 4). Therefore, the shallow layer of SwinS may not bring good enough feature extraction and clustering results. All in all, high-resolution features that describe details are given a higher weight, which helps to locate boundaries and identify small objects; high-level semantic features with higher resolution are also given a higher weight, which helps to obtain better spatial clustering results.

D. Experiments About Boundary Detection

In Section V-A, we explained that the boundary detection strategy based on the CNN model may not be suitable for the current transformer-based backbone. According to the visual analysis of the feature maps, we suspect that feature maps of Stage 3 may be more suitable for boundary detection than Stage 4 in the transformer-based network. In order to verify the assumption, in this section, we designed comparative experiments to compare the results of the two strategies: 1) fusing the feature maps of Stages 1, 2, and 4 to predict the boundary pixels and 2) fusing the feature maps of Stages 1–3 to predict the boundary pixels. Experiments are conducted based on the SwinB-CNN framework on both the Potsdam and Vaihingen datasets, and the input image sizes are set to 300 × 300. The quantitative analysis results are shown in Tables VII and VIII, respectively.

Based on the Potsdam dataset, we can see that both boundary detection strategies improve the accuracy of the proposed

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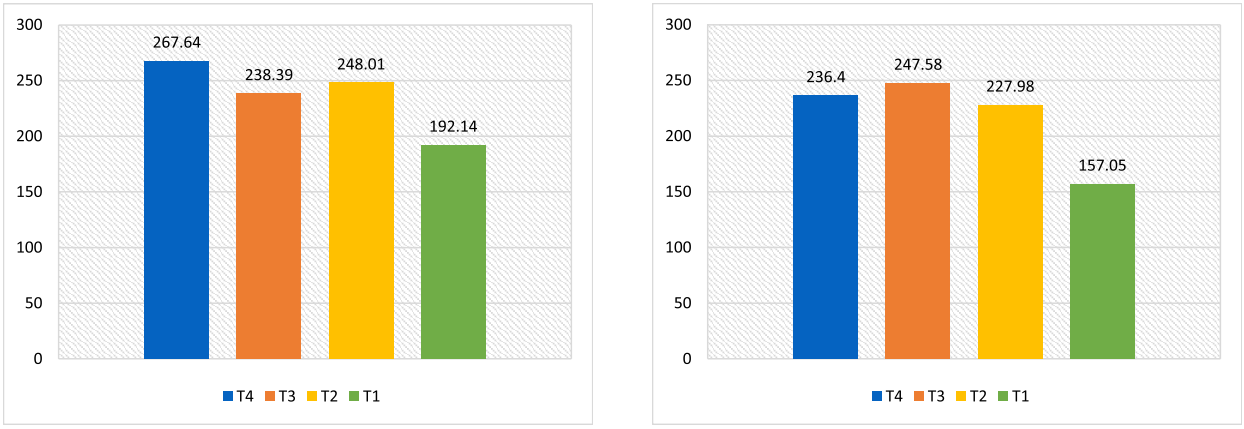


Fig. 10. Weight distribution of the feature maps of different scales calculated by the SE block. The statistics in the figure are the sum of the weights of all feature maps at each scale for (a) SwinB-CNN model and (b) SwinS-CNN model. T_1–T_4 represent the feature maps of different resolutions in the decoder module, T_1 has the lowest spatial resolution, and T_4 has the highest spatial resolution.

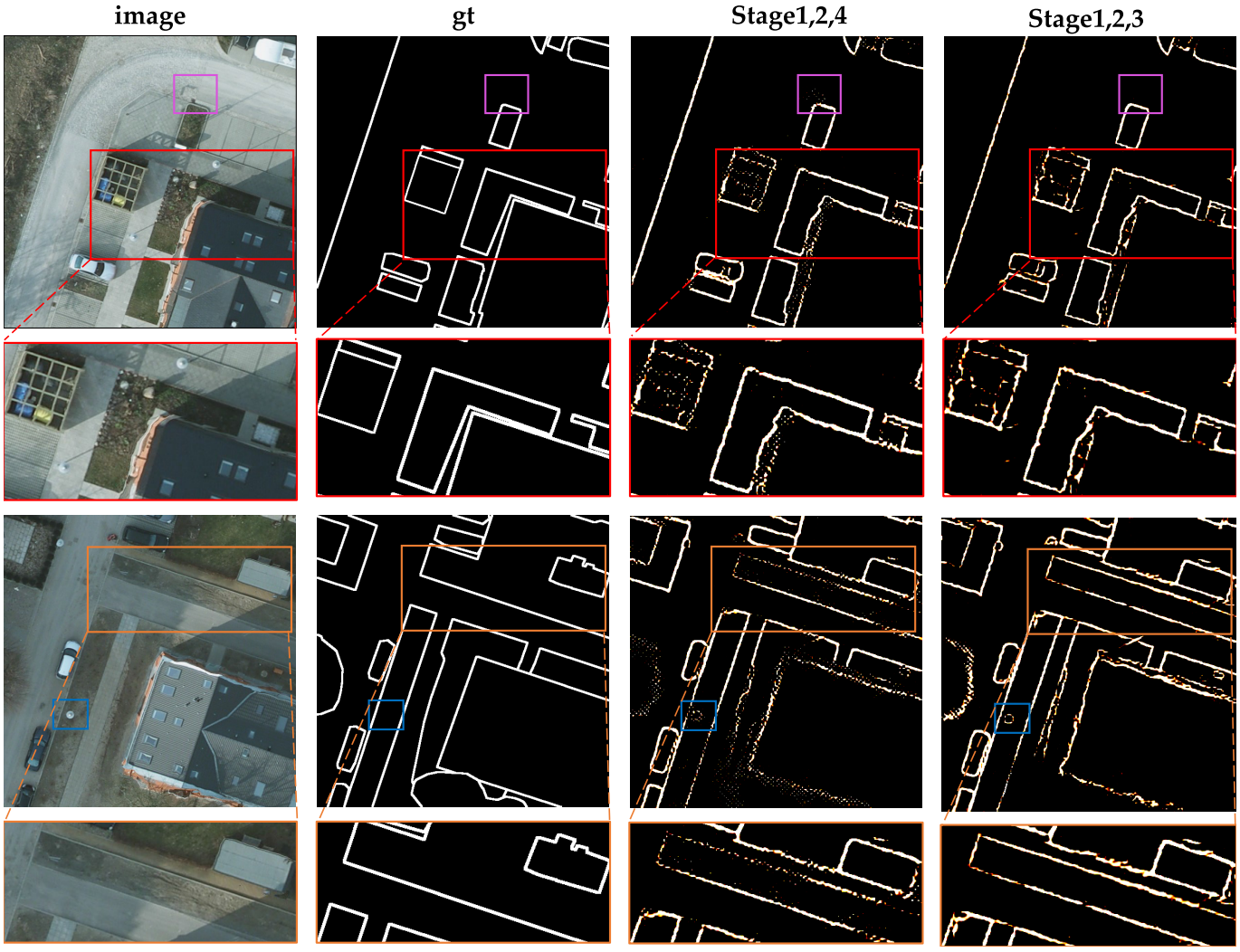


Fig. 11. Comparison of edge detection results based on two different boundary detection strategies on the Potsdam dataset. The first column is the input RGB image, and the second column is the ground truth. The third column is the boundary detection result of the SwinB-CNN + $BD_{1,2,4}$, and the fourth column is the boundary detection result of SwinB-CNN + $BD_{1,2,3}$.

SwinB-CNN model. The first strategy increases the OA of the model by 0.4 percentage points, and the second strategy is 0.7 percentage points. In terms of details, both boundary detection strategies perform better in two classes of building and tree, while the second strategy is more effective in comparison. The results based on the Vaihingen dataset are similar

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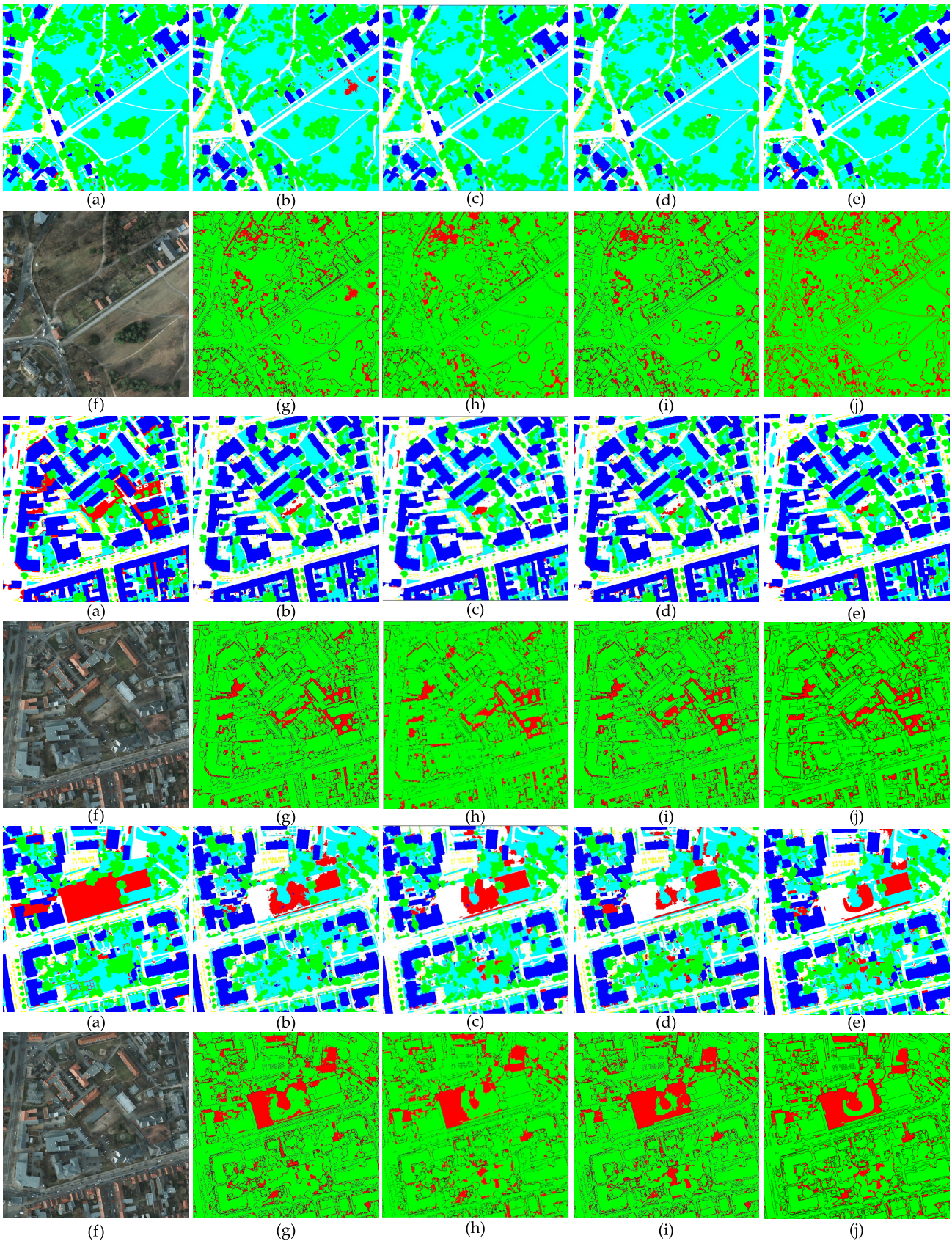


Fig. 12. Comparison between the proposed SwinB-CNN and other state-of-the-art methods. Examples are select randomly from the testing data of the ISPRS Potsdam dataset. Every two rows are a group; from left to right and from top to bottom are (a) ground truth, the results of (b) AFNet + TTA, (c) ResUNet, (d) CASIA2, (e) proposed †SwinB-CNN + $BD_{1,2,3}$, (f) RGB image, the wrongly classified image of (g) AFNet + TTA, (h) ResUNet, (i) CASIA2, and (j) proposed †SwinB-CNN + $BD_{1,2,3}$, respectively.

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TABLE IX

CONFUSION MATRIX OF THE PROPOSED TRANSFORMER AND CNN HYBRID NETWORK (†SWINB-CNN + $BD_{1,2,3}$) ON THE POTSDAM BENCHMARK, OA = 91.5%. THE INPUT IMAGE SIZE IS SET TO 300 × 300, AND THE RESULTS ARE EVALUATED USING AN EDGE-ERODED REFERENCE

↓ predicted ‖ reference →	Building	Car	Low_veg	Imp.	Tree	Background
Building	0.978	0.000	0.003	0.010	0.003	0.006
Car	0.002	0.976	0.000	0.010	0.006	0.006
Low_veg	0.005	0.000	0.903	0.024	0.056	0.012
Imp.	0.007	0.000	0.021	0.952	0.012	0.008
Tree	0.005	0.001	0.093	0.025	0.872	0.005
Background	0.082	0.010	0.122	0.269	0.028	0.489
Precision	0.970	0.952	0.868	0.928	0.896	0.740
Recall	0.978	0.976	0.903	0.952	0.872	0.489
F1 score	0.974	0.964	0.885	0.940	0.883	0.588

TABLE X

CONFUSION MATRIX OF THE PROPOSED TRANSFORMER AND CNN HYBRID NETWORK (†SWINB-CNN + $BD_{1,2,3}$) ON THE VAIHINGEN BENCHMARK, OA = 91.4%. THE INPUT IMAGE SIZE IS SET TO 300 × 300, AND THE RESULTS ARE EVALUATED USING AN EDGE-ERODED REFERENCE

↓ predicted ‖ reference →	Building	Car	Low_veg	Imp.	Tree	Background
Building	0.968	0.001	0.006	0.023	0.002	0.000
Car	0.013	0.905	0.003	0.078	0.001	0.000
Low_veg	0.019	0.000	0.836	0.031	0.113	0.001
Imp.	0.027	0.003	0.034	0.928	0.007	0.001
Tree	0.003	0.000	0.067	0.009	0.921	0.000
Background	0.095	0.042	0.182	0.176	0.010	0.494
Precision	0.952	0.863	0.865	0.938	0.887	0.904
Recall	0.968	0.905	0.836	0.928	0.921	0.494
F1 score	0.960	0.883	0.850	0.933	0.904	0.639

to the results on the Potsdam dataset. Both boundary detection strategies improve the performance of the SwinB-CNN model, and the second strategy is more effective than the first from both the overall and detailed perspectives.

In order to further compare the effectiveness of the two strategies, we visualized some examples of the edge extraction results from the Potsdam dataset in Fig. 11.

From the entire scene (the first and the third rows), it can be seen that the original strategy produces widely distributed salt and pepper noise, and the continuity of the boundary pixels is poor. Some obvious examples are marked in colored rectangles. Since the salt and pepper noise is not displayed clearly in the global scene, local images are enlarged in the second and fourth rows. This supports our assumption that the feature map of the third stage in the Swin transformer backbone is more suitable for edge extraction than the feature map of the fourth stage, so we adopted the second strategy in the final pipeline of the proposed transformer and the CNN hybrid network.

More visual examples of the comparison of the semantic segmentation results obtained by the baseline of SwinB-CNN and adding different boundary detection strategies to the baseline are provided in Appendix C for reference.

E. Results and Comparison With Other State-of-the-Art Methods

In order to further prove the effectiveness of the proposed transformer and the CNN hybrid semantic segmentation network, we add the boundary detection task and the SE block to the "SwinB + 3skips" baseline at the same time and crop the input RGB images with a size of 300 × 300 to train the model. We did not use the maximum size of 448 for consideration of training time and memory consumption. Experiments were conducted on both the Potsdam and Vaihingen datasets. For a fair comparison, we used exactly the same training and evaluation dataset as other participants. In the series of experimental analyses on the proposed architecture, we did not apply the nDSM data for consideration of the generality of the model, while, in experiments in this section, we stacked the nDSM and the optical images as input channels to pursue more competitive semantic segmentation accuracy, and we denote the results as †SwinB-CNN + $BD_{1,2,3}$. Please note that, except for CASIA2, the state-of-the-art models in published papers all use the nDSM data for model training.

Confusion matrices for †SwinB-CNN + $BD_{1,2,3}$ based on the Potsdam and Vaihingen datasets are shown in Tables IX and X, respectively. In general, our model performs very well on both datasets. Based on the Potsdam dataset, we obtained an OA of 91.5%. The F1 scores of building, car, and impervious exceed or approach 95%, and the F1 scores of low vegetation and tree are about 88%. Based on the Vaihingen dataset, our OA is 91.4%. In terms of details, the F1 score of the building exceeds 95%, and the F1 scores of impervious surface and tree exceed 90%.

Furthermore, comparisons based on the Potsdam and Vaihingen datasets between the proposed model and the state-of-the-art methods in the recently published paper are provided in Tables XI and XII, respectively. Both the results of using only the three-band optical imagery

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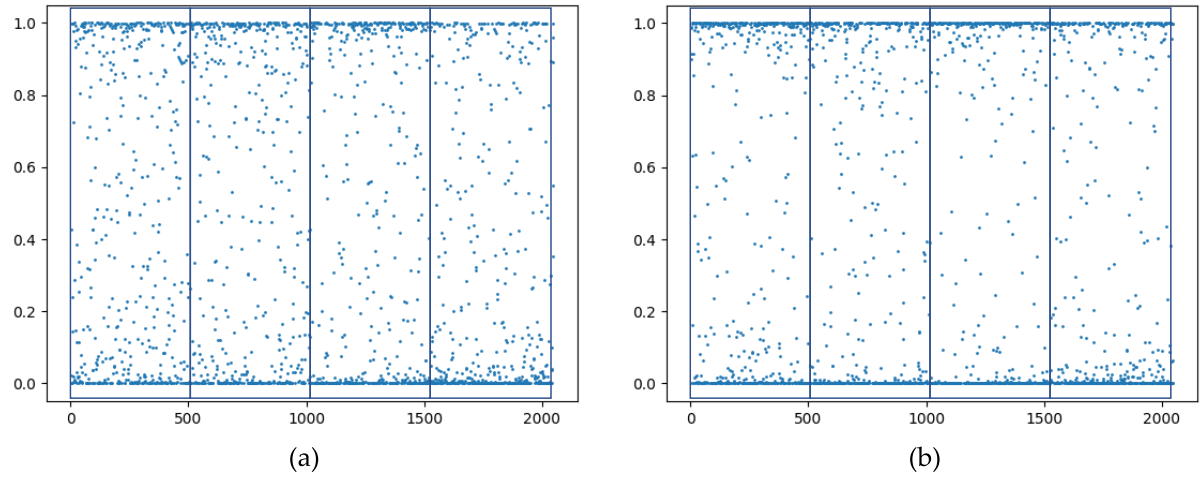


Fig. 13. Weights of channels calculated by the SE block in (a) SwinB-CNN and (b) SwinS-CNN models. Vertical lines separate the feature maps of different scales.

TABLE XI

PERFORMANCE COMPARISON (F1 SCORE AND OA, %) WITH STATE-OF-THE-ART DEEP LEARNING MODELS ON THE POTSDAM BENCHMARK USING AN EDGE-ERODED REFERENCE. THE INPUT IMAGE SIZE IS SET TO 300 × 300. †SWINB-CNN + $BD_{1,2,3}$ REPRESENTS THE RESULTS OF USING THE ADDITIONAL NDSM BAND, AND SWINB-CNN + $BD_{1,2,3}$ REPRESENTS THE RESULTS OF USING ONLY THE THREE-BAND OPTICAL IMAGERY. THE BEST RESULTS ARE MARKED IN BOLD, THE SECOND BEST RESULTS ARE MARKED UNDERLINED, AND THE THIRD BEST RESULTS ARE ENCLOSED IN SQUARE BRACKETS

	Building	Car	Low_veg	Imp.	Tree	OA
UZ_1 [34]	95.4	86.5	81.8	89.3	80.5	85.8
RIT_4 [51]	97.0	95.2	86.9	92.6	87.4	90.3
CASIA2 [36]	97.0	96.2	87.7	93.3	[88.4]	91.1
DST_5 [52]	96.4	94.7	86.7	92.5	88.0	90.3
TreeUNet [35]	[97.3]	95.8	86.8	93.1	87.1	90.7
RS+Edge+DNL [7]	97.1	96.3	87.9	94.0	87.6	91.0
ResUNet [28]	97.2	96.4	[88.2]	93.5	89.2	91.5
AFNet+TTA [37]	**97.6**	**97.1**	**88.7**	**94.1**	**89.7**	**92.1**
†SwinB-CNN+$BD_{1,2,3}$	97.4	96.4	88.5	94.0	88.3	91.5
SwinB-CNN+$BD_{1,2,3}$	96.8	96.3	88.0	93.6	88.0	91.0

TABLE XII

PERFORMANCE COMPARISON (F1 SCORE AND OA, %) WITH STATE-OF-THE-ART DEEP LEARNING MODELS ON THE VAIHINGEN BENCHMARK USING AN EDGE-ERODED REFERENCE. THE INPUT IMAGE SIZE IS SET TO 300 × 300. †SWINB-CNN + $BD_{1,2,3}$ REPRESENTS THE RESULTS OF USING THE ADDITIONAL NDSM BAND, AND SWINB-CNN + $BD_{1,2,3}$ REPRESENTS THE RESULTS OF USING ONLY THE THREE-BAND OPTICAL IMAGERY. THE BEST RESULTS ARE MARKED IN BOLD, THE SECOND BEST RESULTS ARE MARKED UNDERLINED, AND THE THIRD BEST RESULTS ARE ENCLOSED IN SQUARE BRACKETS

	Building	Car	Low_veg	Imp.	Tree	OA
UZ_1 [34]	92.5	57.3	81.6	89.2	86.9	87.3
GSN_3 [53]	95.2	79.3	84.1	92.3	90.0	90.3
DLR [31]	95.2	81.2	83.9	92.4	89.9	90.3
DFN [54]	95.3	[87.2]	83.6	92.6	89.3	90.4
DST_2 [52]	93.7	72.6	83.4	90.5	89.2	89.1
UFMG_4 [9]	94.5	81.3	82.9	91.1	88.8	89.4
CASIA2 [36]	96.0	86.7	[84.7]	93.2	[89.9]	[91.1]
AFNet+TTA [37]	**96.5**	**88.8**	**85.8**	[93.1]	**90.6**	**91.7**
†SwinB-CNN+$BD_{1,2,3}$	96.0	88.3	85.0	**93.3**	90.4	91.4
SwinB-CNN+$BD_{1,2,3}$	95.3	86.9	83.6	92.2	89.6	90.4

(SwinB-CNN + $BD_{1,2,3}$) and the results of adding the nDSM channel (†SwinB-CNN + $BD_{1,2,3}$) are provided in the tables. The proposed †SwinB-CNN + $BD_{1,2,3}$ is competitive with the current state-of-the-art semantic segmentation methods. It achieves the second highest OA on both the Potsdam and Vaihingen datasets, which are only 0.6% and 0.3% lower than the state-of-the-art method "AFNet + TTA" that we know so far. We guess that the reason for the higher accuracy of the AFNet + TTA model is that it uses two backbones to process the optical and the nDSM/normalized difference vegetation index (NDVI) data separately. However, this processing method will introduce more model parameters, and when height information is not available, the versatility of the model will be affected.

In terms of details, on the Potsdam dataset, our model ranks second in the F1 score for all subclasses except trees. On the Vaihingen dataset, our model ranks first in the F1 score of the impervious class and ranks second in the F1 scores of other subclasses. It is worth noting that, without adding the nDSM data, the proposed model has achieved an OA of 91.0% on the Potsdam dataset and 90.4% on the Vaihingen dataset. It shows the good versatility of the proposed model.

A visual comparison between the results of our proposed model and the results of the three state-of-the-art methods based on the Potsdam dataset are shown in Fig. 12. Since the ISPRS has stopped completing the accuracy assessment for participants and provided all ground truth in 2018, we wrote code and completed the accuracy assessment honestly. The authors of "AFNet + TTA" give a link of their results in the published paper, the results of ResUNet were cropped from the paper, and the results of CASIA2 were downloaded from the website, so the color and presentation of these results look slightly different. However, this does not affect one's comparison of them at all.

It can be seen that the proposed †SwinB-CNN + $BD_{1,2,3}$ performs well on the Potsdam dataset. There are few misclas-

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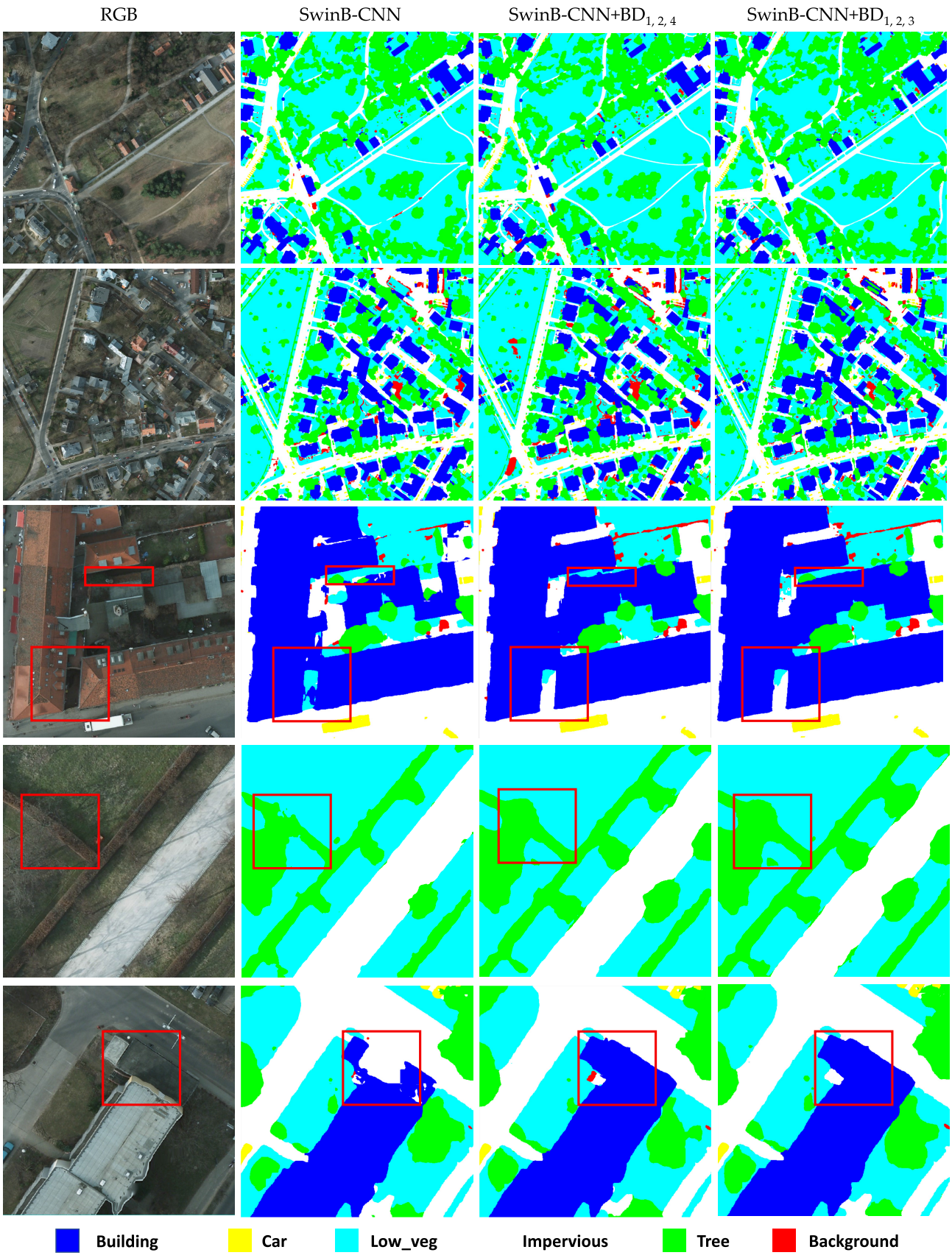


Fig. 14. Comparison of semantic segmentation results obtained by the SwinB-CNN baseline or add different boundary detection branches to it. The red boxes mark some area where the semantic segmentation result is obviously optimized.

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TABLE XIII
COMPARISON OF OA (%) BETWEEN THE STATE-OF-THE-ART BACKBONES BASED ON AIRS BENCHMARK

BackBone	Building	Non-building	OA
ResNeSt-101 [22]	92.7	99.51	98.47
ResNet-101 [13]	93.35	99.43	98.50
HRNet-W48 [14]	93.38	99.06	98.39
ViT [16]	85.90	97.66	95.55
SegFormer-B4 [42]	93.60	98.58	96.09
SwinB [20]	94.69	99.21	98.41

sified pixels, most of them appear near the edge area, and a small part of misclassification is caused by too large intraclass variance. It is difficult for us to distinguish which algorithm is better through simple observation of these examples, but it can be seen that †SwinB-CNN + $BD_{1,2,3}$ has strong competitiveness compared with other state-of-the-art algorithms. This proves the effectiveness of our algorithm.

VI. CONCLUSION

In this work, we propose a novel transformer and CNN hybrid deep learning network for semantic segmentation of VHR remote sensing images. The model uses an encoder–decoder structure. The encoder module applies a Swin transformer backbone. Compared with the CNN-based backbone, the Swin transformer does not include inductive bias and thus can better express long-distance dependence in spatial relationships. Compared with other transformer-based backbones, the advantages of the Swin transformer backbone includes low computational complexity, fewer parameters, and output of hierarchical feature maps.

The decoder module draws on multiple effective blocks and successful strategies in the study of semantic segmentation of remote sensing images based on CNN. In the middle part of the network, the SASPP block is used to extract multiscale context information. The U-shaped decoder is used to gradually restore the size of the feature map, and three skip connections between deep and shallow feature maps are established to maintain the transmission of local details and enhance the communication of multiscale features. The SE channel attention block is inserted to increase the weight of the feature map that is more important to the semantic segmentation for feature augmentation. A dual-task design is applied; the boundary detection branch is added to reduce the salt and pepper noise, and improve the accuracy of boundary positioning in the semantic segmentation results.

Extensive experiments are designed to analyze the advantages of our proposed algorithm based on the Potsdam and Vaihingen benchmarks. We first compared the feature extraction capabilities, parameter sizes, and computation complexity of the most popular CNN- and transformer-based backbones. Then, we visualized and analyzed the feature maps of SwinB and ResNet backbones. Second, we tested the effectiveness of the skip connections and the SE block in our proposed decoder module. Subsequently, we compared two different edge detection strategies and proved the rationality of our boundary extraction strategy designed for the Swin transformer backbone. Finally, we compared the results of our proposed network with the published state-of-the-art methods on both the Potsdam and Vaihingen benchmarks. The proposed model achieves the second highest OA on both datasets and also performs well in all five subclasses.

The work of this article is an attempt to use a hybrid model of transformer and CNN for semantic segmentation of remote sensing images. Some latest works on semantic segmentation based on transformer and CNN are merged. Although good semantic segmentation results have been achieved, the parameters of the model are relatively large, and the network structure design of the boundary detection task is still simple. Our future work needs to try to simplify the network structure and design a more effective boundary detection branch to improve the semantic segmentation results.

APPENDIX A

AIRS is a new large-scale benchmark dataset termed aerial imagery for roof segmentation. The AIRS dataset covers almost the entire Christchurch City in New Zealand with about 457 km^2. It contains over 220 000 buildings, and the ground sampling distance of the AIRS dataset is 7.5 cm. AIRS contains 951 image blocks, of which 857 are used for training, and the remaining 94 blocks are used for testing. The size of each image block is 10 000 × 10 000 pixels. Two classes of buildings and nonbuildings need to be labeled.

Our goal of this set of experiments is to test whether SegFormer and ViT perform better on larger datasets. Only two land cover types of buildings and nonbuildings need to be labeled, and the nonbuilding pixels account for a large proportion. When only considering the OA, the performance of multiple deep neural networks on the AIRS dataset is indistinguishable. However, one can still see that SegFormer and ViT still have a small gap with the other popular backbones on the large-scale dataset.

APPENDIX B

Fig. 13 shows the scatter plots of the weight of each channel calculated by the SE block in the SwinB-CNN and SwinS-CNN models based on the Potsdam benchmark.

APPENDIX C

Fig. 14 shows the comparison of the semantic segmentation results of the original SwinB-CNN by adding different boundary detection strategies to the SwinB-CNN. As shown in the figure, the edge constraint improves the visual effect of the semantic segmentation result very obviously. Some salt and pepper noises in the original semantic segmentation results are removed, the structure of the targets is more complete, and the predicted edge pixels fit the real boundaries more closely. For the two boundary detection strategies, the result of the SwinB-CNN + $BD_{1,2,3}$ model is better than that of the SwinB-CNN + $BD_{1,2,SASPP}$.

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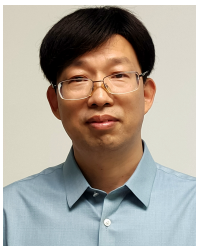
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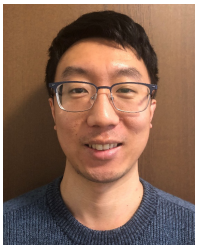
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